Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/18/2015
PI Name:	Perera, Imara Y Ph.D.		
Project Title:	The Effect of Spaceflight on Transgenic Arabidopsis Plants With Compromised Signaling		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27695	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	07/01/2010	**End Date:**	06/30/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 ; ISS NOTE: Extended to 6/30/2015 per NSSC (Ed., 5/5/14) NOTE: Extended to 6/30/2014 per A. Chu/ARC (Ed., 5/28/2013)		
Key Personnel Changes/Previous PI:	Caroline M Smith - Research Associate ; Eric S Land- Research Assistant		
COI Name (Institution):	Brown, Christopher (North Carolina State University) Sederoff, Heike (North Carolina State University)		
Grant/Contract No.:	NNX10AM72G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants provide a complete and economical means for human life support for long-term space exploration and habitation. However, since the space environment is not optimal for plant growth, an understanding of how plants sense and respond to changes in their environment is of fundamental importance. The phosphoinositide (PI) pathway is highly conserved among eukaryotes and functions in the regulation of a multitude of cellular pathways. The lipid-derived second messenger inositol 1,4,5-trisphosphate (InsP3) increases in response to many different stresses. We have shown that InsP3 levels increase with gravistimulation prior to visible bending in both monocot and dicot systems. We have generated transgenic Arabidopsis plants expressing the mammalian type I inositol polyphosphate 5-phosphatase (InsP 5-ptase), an enzyme that specifically hydrolyzes InsP3 and terminates the signal. The transgenic plants have normal growth and morphology; however, they exhibit altered responses to many environmental stimuli including gravity, drought, and cold. While rapid changes in transcript levels occur in wild type Arabidopsis within 5 min of gravistimulation, the expression of several of the fastest responding genes does not change in the InsP 5-ptase roots in response to reorientation. Our hypothesis is that InsP3 is an important second messenger in the sensing and signaling of stimuli (including gravity). The plants with compromised InsP3 signaling therefore provide a valuable tool for dissecting the role of the InsP3 pathway in plant responses to the microgravity environment encountered on the International Space Station. Our primary aim is to identify the molecular changes that are specifically mediated by InsP3 in the space environment. We will compare transcript and protein profiles of wild type and transgenic plants that are grown in both microgravity and 1g in space as well as on the ground. The long term goal is to understand the molecular mechanisms plants use to sense and respond to changes in their environment. This knowledge will help design plants which are better able to withstand space flight and microgravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The InsP 5-ptase transgenic plants provide a good model system for understanding several issues including plant responses to gravity, plant responses to environmental stress, and fundamental aspects of PI signaling. Our previous work has implicated InsP3 as a key player in gravity sensing and signaling and the transgenic plants have attenuated InsP3 signaling and reduced gravitropism. In addition these plants have altered responses to different environmental stresses. Therefore this study will help us understand plant signaling during environmental stress, which has long term implications with improving crops to better withstand unfavorable conditions. Furthermore, the phosphoinositide pathway is highly conserved among plants, yeast, and mammals. These studies using the transgenic plants with attenuated InsP3 signaling will provide valuable insights on the regulation and downstream effectors of this pathway and could also contribute to our understanding of the role of the PI pathway in regulating human physiology.
Task Progress:	We have carried out a second RNA-Sequencing experiment on the Illumina platform utilizing root samples from position 3 cassettes. We are currently analyzing/processing the data and will compare the differential gene expression results from both RNA-Seq trials. We have also set up collaborations with colleagues in the College to further analyze the results. We will determine the effect of the fractional g gradient by comparing the results from the different cassette positions. We are also looking to develop a gene regulatory network for root development under microgravity. The follow-up proposal for flight experiments in response to the NASA solicitation NNH14ZTT001N submitted in March 2014 was selected for funding and is entitled, "Transcriptional and Post Transcriptional Regulation of Seedling Development in Microgravity." Presentations: Poster presentation- Transcriptional Regulation of seedling development in microgravity. Annual meeting of the American Society of Plant Biologists 2015.
Bibliography Type:	Description: (Last Updated: 05/30/2025)
Articles in Peer-reviewed Journals	Dalal J, Land E, Vasani N, He L, Smith C, Rodriguez-Welsh M, Perera IY, Sederoff H. "Methods for RNA profiling of gravi-responding plant tissues." Methods Mol Biol. 2015;1309:91-117. (In: Plant Gravitropism: Methods and Protocols. E.B. Blancaflor, editor. New York: Springer, 2015.) http://dx.doi.org/10.1007/978-1-4939-2697-8_9 ; PubMed PMID: 25981771 , May-2015
Articles in Peer-reviewed Journals	Sheppard J, Land ES, Toennisson TA, Doherty CJ, Perera IY. "Uncovering transcriptional responses to fractional gravity in Arabidopsis roots." Life (Basel). 2021 Oct;11(10):1010. https://doi.org/10.3390/life11101010 ; PMID: 34685382; PMCID: PMC8539686 , Oct-2021
Articles in Peer-reviewed Journals	Land ES, Sheppard J, Doherty CJ, Perera IY. "Conserved plant transcriptional responses to microgravity from two consecutive spaceflight experiments." Front Plant Sci. 2024 Jan 8;14:1308713. https://doi.org/10.3389/fpls.2023.1308713 ; PMID: 38259952; PMCID: PMC10800490 , Jan-2024